1	The table below shows the estimates of some physical quantities. Which row is **not** a reasonable estimate?

		quantity	value
	A	current used in an electric kettle	8 A
	B	mass of an adult male	70 kg
	C	speed of an Olympic sprint runner	10 m s^{-1}
	D	water pressure at the bottom of a public swimming pool	10^4 Pa

Answer: D

Option A: With I = 8 A, and V = 240 V, power rating of kettle = 8 x 240 = 1920 W which is a reasonable estimate.

Option B: Typically Singaporean adult weighs between 50 kg and 100 kg. Therefore 70 kg is a reasonable estimate.

Option C: The timing for 100 m race is the Olympic is about 10 s; therefore the average speed of 10 m s^{-1} is a reasonable estimate.

Option D: Atmospheric pressure = 10^5 Pa.
Assume that the swimming pool has depth of 2 m,
Pressure at the bottom pool = Atmospheric pressure + hρg
= 10^5 + (2 x 1000 x 9.81)
= 10^5 + 1.96 x 10^4 = 1.196 x 10^5 Pa

10^4 Pa is less than 1.196 x 10^5 Pa. Therefore 10^4 Pa is <u>not</u> a reasonable estimate. Answer is D.